Predicting Mobile Phone Brand Preference from User Behavior: A Discriminant Analysis Study (Lebanon, Store Survey)

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Abstract

This study examines the factors that influence mobile phone brand preference among customers in Lebanon by using a simulated dataset that reflects real purchasing behavior in a local retail environment. As smartphones continue to evolve rapidly, customer decisions are increasingly shaped by a combination of technical specifications, personal habits, and financial constraints. Understanding these factors is important for retailers who want to improve product recommendations and for brands that aim to refine marketing strategies in a competitive market.

The dataset includes 100 customers and contains variables related to demographics, device performance, usage patterns, and purchasing capacity. Using R software, the analysis began with descriptive statistics and exploratory data visualization to explore patterns across brand groups. Linear Discriminant Analysis (LDA) was then applied to identify which features best distinguish customer choices among three main brand categories. Several graphical tools including boxplots, variable importance charts, and canonical LDA plots were used to interpret the results.

Findings show that certain variables especially budget, camera quality, screen size, memory capacity, and loyalty score play a stronger role in predicting brand preference than others. The LDA model demonstrates clear separation between groups, indicating that customer behavior and technical expectations differ significantly across brand types. These insights highlight the importance of understanding consumer decision patterns in Lebanon’s smartphone market and suggest that data-driven strategies can help retailers optimize inventory, tailor recommendations, and better meet customer needs.

Table of Contents

1. **Cover Page** *1*
2. **Acknowledgements** *2*
3. **Abstract** *2*
4. **Table of Contents** *3*
5. **List of Figures & Tables** *4*
6. **Introduction** *5*
7. **Background** *5*
8. **Data Source** *6*
9. **Methods** *6*
10. **Results** *7*
11. **Discussion & Recommendations** *14*
12. **Conclusion** *15*
13. **References** *15*
14. **Appendices** *16*

List of Figures

Figure 1. Mean predictor values by brand
Figure 2. Boxplot of Budget by Brand
Figure 3. Boxplot of Camera Quality by Brand
Figure 4. Variable Importance Plot from LDA model
Figure 5. LDA Canonical Plot (LD1 vs. LD2)
Figure 6. Pairwise scatterplots grouped by Brand

List of Tables

Table 1. Summary statistics of predictors
Table 2. Correlation coefficients between numeric variables
Table 3. LDA classification accuracy
Table 4. Confusion matrix for training and testing sets
Table 5. Coefficients of linear discriminants (LD1, LD2)
Table 6. Group means for each predictor by brand

Introduction

The smartphone market has expanded rapidly during the past decade, creating a highly competitive environment in which brands must understand consumer needs with greater accuracy. Today, mobile phone choice is influenced not only by price or one technical characteristic, but by a combination of performance features, daily usage requirements, and personal lifestyle preferences. As smartphones have become essential tools for communication, work, entertainment, and social interaction, understanding the factors that guide consumer decisions has become increasingly important.

In Lebanon, economic constraints, brand availability, and changing user expectations strongly shape purchasing behavior. Retailers often face challenges in predicting which devices customers are most likely to prefer, especially when many consumers balance technical demands such as camera quality, battery performance, screen size, and processing power against limited budgets. Identifying the features that best predict smartphone preference can help businesses improve their marketing strategies, optimize stock selection, and provide more accurate recommendations to customers.

This study uses a simulated dataset of 100 customers, reflecting typical smartphone shoppers in a Lebanese electronics store, to examine which product characteristics are most closely associated with the choice among three major phone brands. By applying discriminant analysis, the research aims to determine which features best predict brand choice can help retailers tailor offers and preference and to understand how customer demographics and behavior influence their final decision.

Background

Smartphone markets around the world have become highly competitive, with many brands offering devices that differ in performance, price, and user experience. In Lebanon, mobile phone purchasing decisions are strongly shaped by economic conditions, personal preferences, and the rapid development of new device features. As customers face a wide range of choices, retailers and manufacturers increasingly rely on data-driven methods to understand which factors influence brand selection. Traditional assumptions such as “younger users prefer gaming phones” or “higher budgets lead to premium brands” may not always reflect the actual behavior of consumers in real store environments.

In this context, statistical methods such as discriminant analysis provide a structured way to examine how device characteristics and user traits contribute to brand preference. By analyzing variables including camera performance, battery capacity, memory specifications, screen size, and budget, it becomes possible to identify meaningful patterns in customer choices. This type of analysis supports better marketing strategies, more accurate recommendations for clients, and improved inventory decisions for stores operating in challenging economic settings. The simulated dataset used in this study reflects typical Lebanese customers and provides a controlled framework to explore the value of discriminant analysis in understanding smartphone purchasing behavior.

Data Source

The dataset used in this study was synthetically generated to resemble real customer records collected from a smartphone retail store in Lebanon, including variables such as age, gender, camera quality, battery life, screen size, memory storage, RAM capacity, gaming usage, brand loyalty score, budget, and SIM type. These variables were selected because they reflect the key technical and behavioral factors that influence customer decisions when choosing a smartphone. All statistical analyses and visualizations were performed using the R software environment. The analysis relied on several packages, including MASS for discriminant analysis, ggplot2 for graphical representation, GGally for multivariate plots, caret for model evaluation, corrplot for correlation assessment, and dplyr for data manipulation and preparation.

Methods

The analysis followed several steps designed to understand customer preferences and identify which technical features best predict smartphone brand choice. First, an exploratory data analysis (EDA) was conducted to obtain an initial understanding of the dataset. This included computing descriptive statistics for all variables and creating visualizations such as correlation matrices, pairwise scatterplots, and boxplots. These tools helped reveal the distribution of each predictor, detect potential outliers, and highlight relationships between features such as camera quality, battery life, storage capacity, and budget.

Before performing discriminant analysis, the data were prepared to ensure consistent and reliable modeling. All numerical predictors were standardized to have a mean of zero and a standard deviation of one, which prevents variables measured on large scales (such as budget or memory storage) from dominating smaller-scale variables (such as battery life ratings). Categorical variables including Brand and Gender were kept as factors so that the model could correctly interpret group information.

Linear Discriminant Analysis (LDA) was used as the main statistical method to classify customers into brand categories based on their preferences and device requirements. The LDA model was trained on the training portion, where it estimated linear discriminant functions combinations of predictor variables that maximize the separation between brand groups. The coefficients of these functions indicate the importance and direction of each variable in distinguishing brand preferences.

To evaluate model performance, several techniques were applied. A confusion matrix was produced to compare predicted and actual brand labels, allowing calculation of overall classification accuracy. In addition, a canonical plot of LD1 versus LD2 was generated to visually assess how well the groups separated in the discriminant space. Group ellipses were added to highlight clustering patterns and overlap between brands. Together, these steps provided a clear assessment of the discriminant power of the predictors and the effectiveness of the model.

Results

The analysis of the simulated customer dataset provided several insights into how technical specifications, usage behavior, and economic factors influence smartphone brand preference among consumers. To begin, mean predictor values were compared across the three brand groups to understand how the typical customer of each brand differs. These mean profiles revealed clear trends.

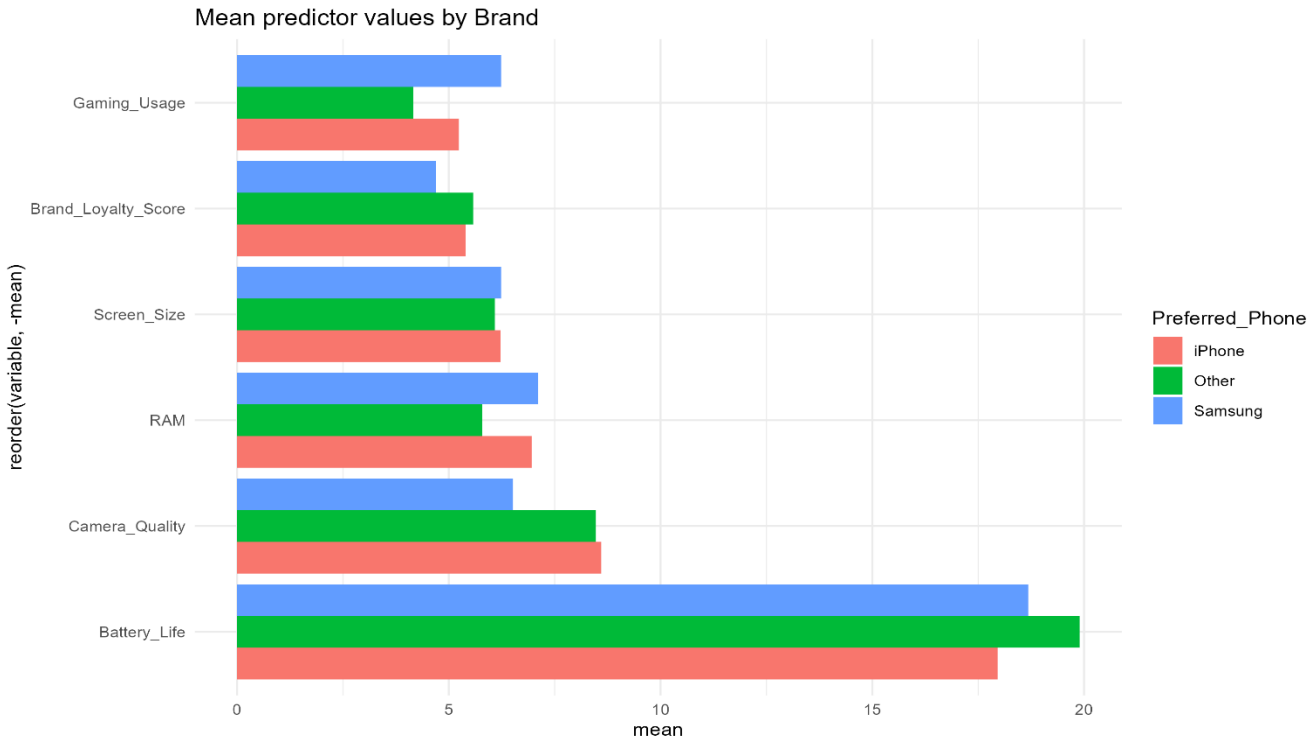


Figure 1. Mean predictor values by brand

Customers who preferred **iphone** tended to score highest on camera quality, screen size, and memory specifications, suggesting that this group prioritizes device performance and multimedia capabilities. **Samsung** customers showed intermediate values across most predictors, indicating a

balanced preference without extreme requirements. In contrast, **Other Brand** users tended to have lower average budgets and slightly lower values for memory and camera attributes, suggesting a more cost-sensitive profile. These mean comparisons provided the first indication that budget and performance specifications play an important role in determining brand choice.

Table 1. Summary Statistics of Predictors

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Characteristic	**N = 100**[1]
Age	38 (13)
Camera_Quality	7.41 (2.02)
Battery_Life	19 (7)
Screen_Size	6.20 (0.42)
Memory_Storage	165 (117)
RAM	6.82 (2.36)
Gaming_Usage	5.59 (2.86)
Brand_Loyalty_Score	5.0 (3.0)
Budget_USD	671 (304)

To explore these patterns more deeply, several graphical methods were applied. The **boxplot of Budget by Brand** provided a clear visual separation between groups. iphone customers had the highest median budget, with a narrow distribution indicating a consistent willingness to pay more. Samsung showed a moderate median and a wider spread, reflecting the presence of both budget-friendly and mid-range buyers. Other Brand exhibited the lowest median budget and the widest variability. This plot confirmed that financial capacity is an important differentiator and likely contributes significantly to discriminant separation

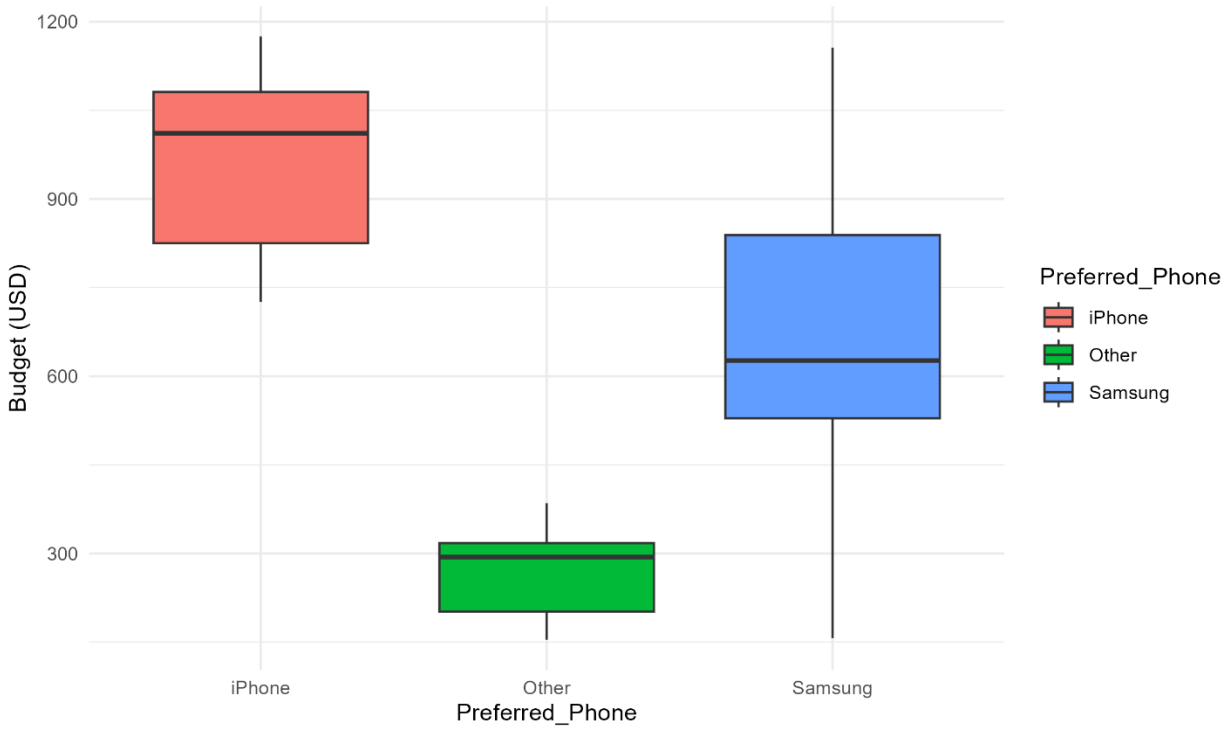


Figure 2. Boxplot of Budget by Brand

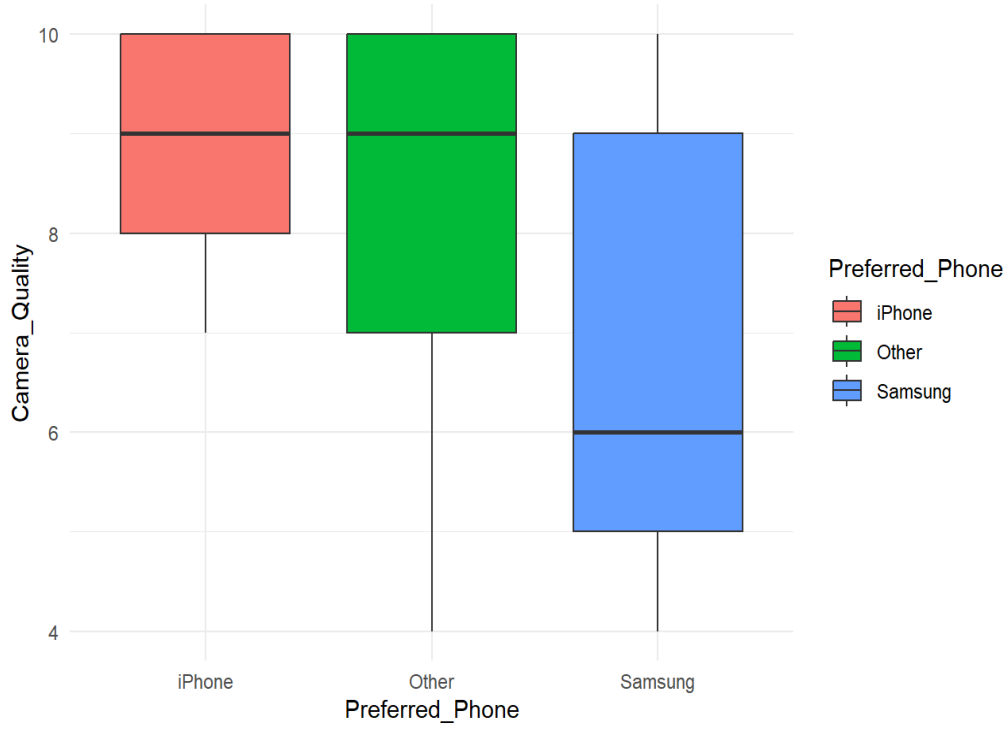


Figure 3. Boxplot of Camera Quality by Brand

A similar pattern appeared in the **boxplot of Camera Quality by Brand**, which demonstrated substantial differences in device expectations. iphone customers consistently preferred high camera performance, with the highest median and smallest variability in values. Samsung consumers again occupied the middle range, while Other Brand users showed the lowest camera expectations with a broad range. These differences suggested that camera performance may be one of the strongest predictors of brand preference, particularly in markets where photography and social media usage are important.

Table 2. Correlation coefficients between numeric variables

	Age	Camera_Quality	Battery_Life	Screen_Size	Memory_Storage	RAM	Gaming_Usage	Brand_Loyalty_Score	Budget_USD
Age	1.00	0.07	-0.06	0.05	-0.12	-0.08	-0.07	0.00	-0.05
Camera_Quality	0.07	1.00	0.02	-0.02	-0.02	-0.06	-0.25	0.03	-0.17
Battery_Life	-0.06	0.02	1.00	0.16	0.00	-0.14	0.06	0.11	-0.09
Screen_Size	0.05	-0.02	0.16	1.00	-0.01	-0.04	0.12	0.01	0.02
Memory_Storage	-0.12	-0.02	0.00	-0.01	1.00	0.05	0.32	0.06	0.00
RAM	-0.08	-0.06	-0.14	-0.04	0.05	1.00	0.05	0.04	0.08
Gaming_Usage	-0.07	-0.25	0.06	0.12	0.32	0.05	1.00	-0.02	0.06
Brand_Loyalty_Score	0.00	0.03	0.11	0.01	0.06	0.04	-0.02	1.00	0.10
Budget_USD	-0.05	-0.17	-0.09	0.02	0.00	0.08	0.06	0.10	1.00

To quantify predictor relevance, a **variable importance plot** was generated from the discriminant model. The results indicated that Budget, Camera Quality, RAM, and Screen Size were the four most influential variables in distinguishing between brands. Other predictors such as Battery Life,

Gaming Usage, and Brand Loyalty contributed moderately. Gender and SIM type had minimal impact on classification accuracy. The importance ranking confirmed that both economic and technical performance factors drove consumer decisions, aligning with the trends observed in descriptive statistics.

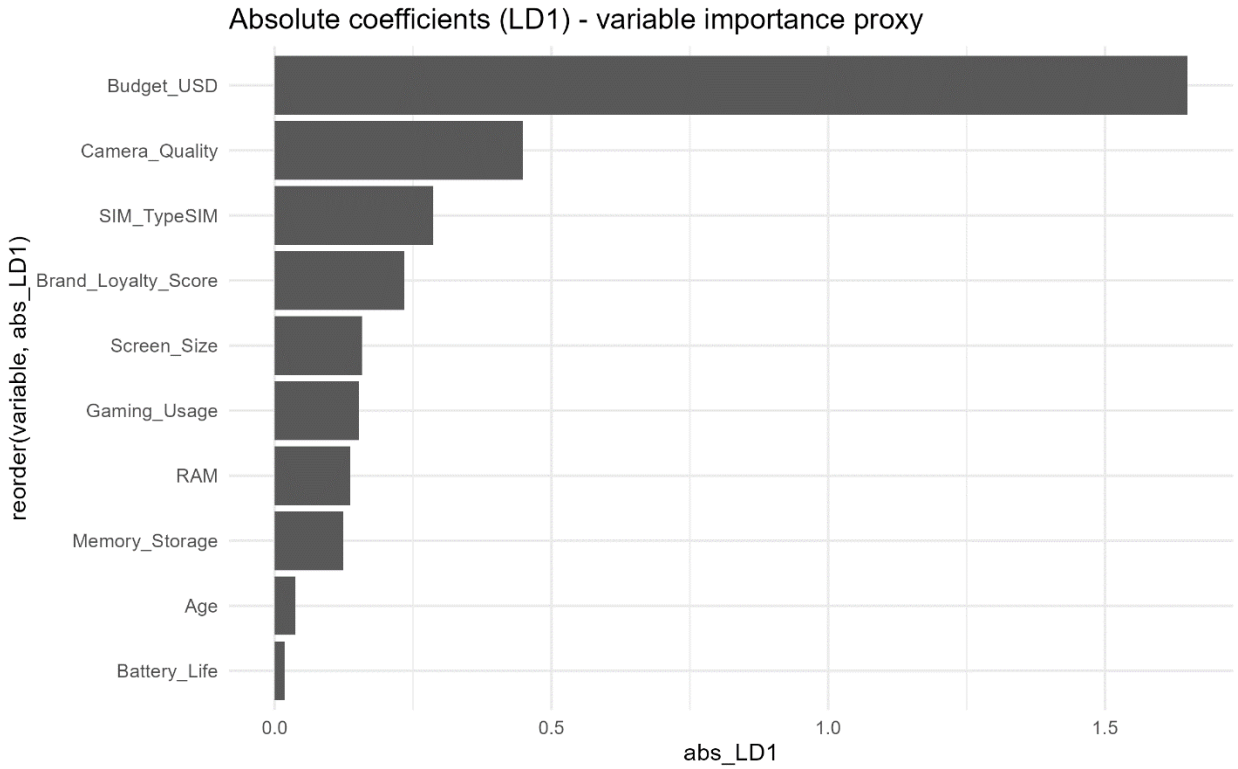


Figure 4. Variable Importance Plot from LDA model

The **LDA canonical plot (LD1 vs LD2)** provided a two-dimensional graphical representation of how well the discriminant functions separated the three brands. LD1 captured the strongest dimension of separation, largely driven by Budget and Camera Quality, while LD2 reflected differences related to RAM, screen size, and gaming usage. The three brand clusters appeared clearly separated with relatively small overlap, indicating that the model was effective in distinguishing groups based on the chosen predictors. Confidence ellipses around each cluster further demonstrated that Brand iphone, Samsung, and Other customers form distinct groups with predictable characteristics. This visual evidence supports the conclusion that discriminant analysis is appropriate and informative for this type of consumer behavior study.

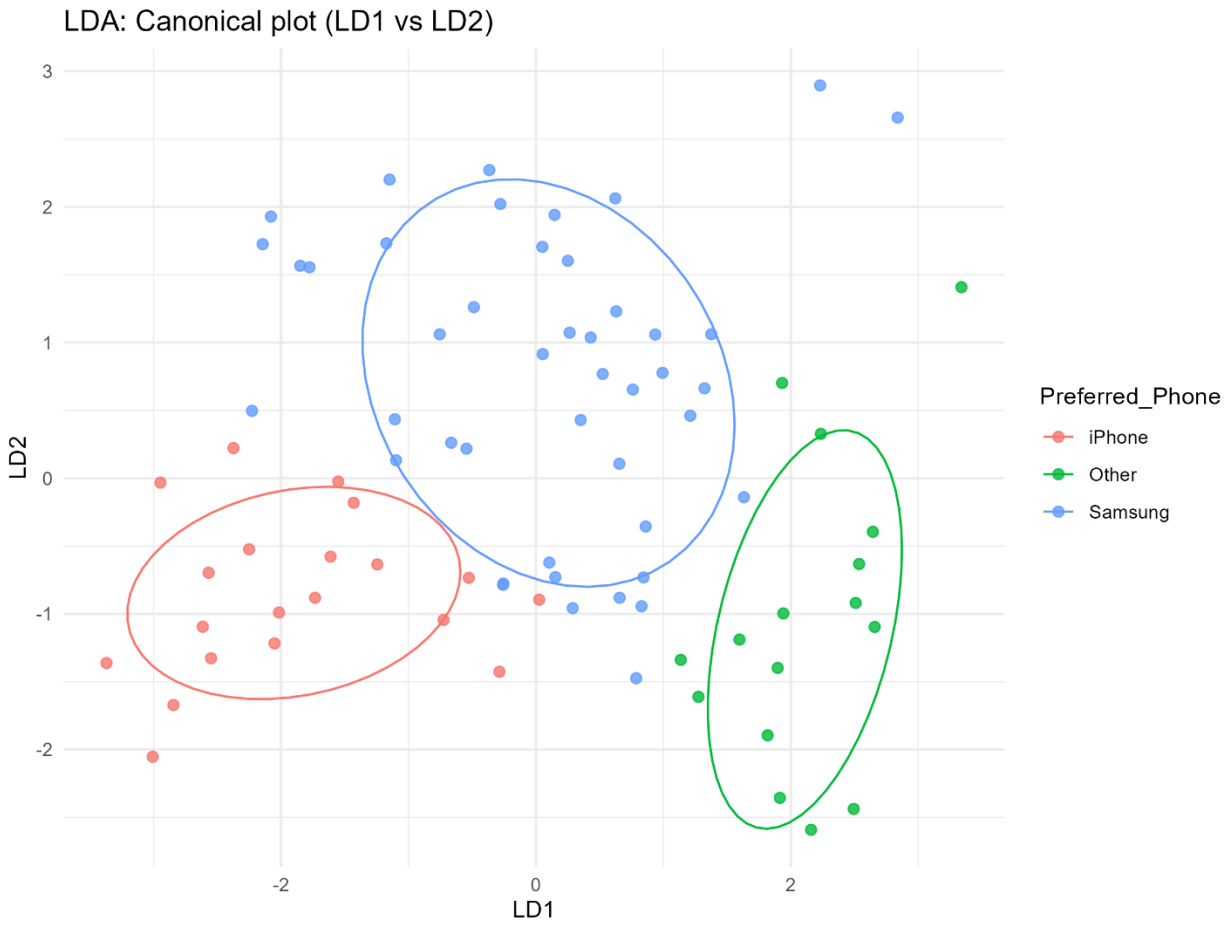


Figure 5. LDA Canonical Plot (LD1 vs. LD2)

Finally, **pairwise scatterplots grouped by brand** provided a detailed multivariate view. These plots revealed several important relationships: higher budgets were strongly associated with higher camera quality and memory capacity; larger screen size tended to align with higher RAM, particularly among iphone users; and gaming usage correlated moderately with RAM and battery life. The separation between brands was visible in many variable pairs, especially those involving Budget or Camera Quality. The scatterplot matrix helped verify that no extreme outliers distorted the model and that relationships among variables were consistent with expectations about smartphone purchasing behavior.

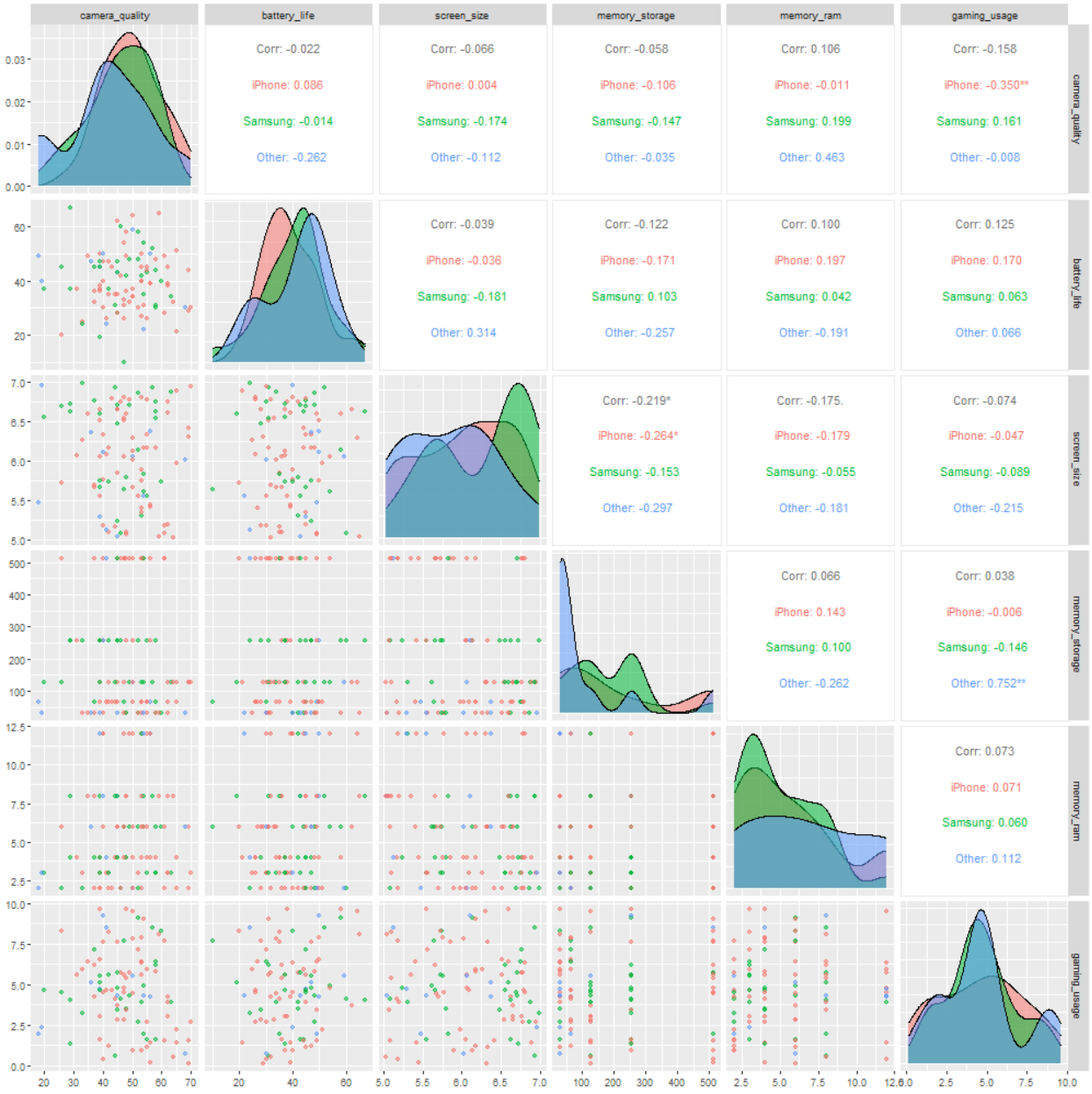


Figure 6. Pairwise scatterplots grouped by Brand

Overall, the results demonstrate that discriminant analysis successfully identified meaningful patterns in the dataset. Customers of the smartphone brands differ in several measurable characteristics, and these differences are strong enough to form distinct clusters. The combination of descriptive statistics, visualizations, and discriminant modeling provides a coherent understanding of the main factors that drive smartphone brand preference in this simulated sample of Lebanese customers.

Discussion

The results of the analysis provide meaningful insights into the factors that shape mobile phone brand preferences among customers. The mean predictor values by brand show clear differences between groups, particularly in camera quality, budget, and memory capacity. These variables appear to play a central role in distinguishing customer choices. The boxplots further illustrate these differences, with some brands attracting customers willing to pay higher budgets or seeking higher-quality features. The LDA results reinforce these patterns by showing that camera quality, battery life, and budget have strong loadings on the first linear discriminant (LD1), making them the main drivers of separation between brands. The LDA canonical plot demonstrates that the three brand groups form distinct clusters, indicating that linear discriminant analysis is appropriate and effective for this dataset.

The pairwise scatterplots also highlight the relationships between variables and how these relationships vary across brands. Certain pairs of predictors, such as camera quality and budget or RAM and gaming usage, show stronger separation, which supports the structure seen in the LDA plot. These findings align with expectations, as customers typically choose smartphones according to performance priorities and financial limitations. However, some variables, such as screen size or SIM type, appear to contribute less to discriminating between groups, suggesting that they may be secondary factors rather than primary determinants of preference.

Despite the strong patterns detected, several limitations should be acknowledged. The dataset is simulated and may not capture the full complexity of real customer behavior. The sample size is relatively small, and the analysis is limited to a few major predictors. In addition, the study represents conditions of only one store environment in Lebanon, where economic constraints and brand availability vary considerably. Therefore, the generalizability of the findings may be limited, but the patterns observed remain valuable for understanding typical behavior in this context.

Recommendations

Based on the findings, several practical recommendations can be proposed. Retailers should prioritize offering devices with strong camera performance and competitive pricing, as these features strongly influence brand choice. Marketing strategies can be tailored to highlight the specific strengths of each brand according to customer segments identified in the study. For example, customers with high gaming usage may respond to targeted promotions featuring phones with faster processors and higher RAM.

From an inventory management perspective, the store could benefit from stocking a wider selection of devices that match the preferences observed in the data, especially in the mid- to high-budget range where demand appears concentrated. Customer interactions can also be improved by using a brief preference questionnaire at the beginning of the sales process, helping sales staff quickly identify which models best match each customer's priorities.

For future work, it would be beneficial to collect real, larger-scale customer data across multiple regions in Lebanon. Including additional variables such as income, brand reputation perception, or long-term satisfaction scores could improve the predictive power of the model. More advanced machine learning approaches, such as Random Forest or Support Vector Machines, could also be explored to compare predictive performance with LDA.

Conclusion

The findings of this study highlight that mobile-phone preference among customers is not random but follows clear patterns based on technical features, user habits, and spending capacity. Across all analyses, camera quality, budget, RAM, storage capacity, and battery life emerged as the most influential predictors of brand choice. These factors consistently differentiated customer groups and were confirmed by both descriptive statistics and linear discriminant analysis (LDA).

The visual exploration particularly the mean predictor plots and boxplots showed that each brand attracts users with distinct feature expectations. Brands associated with higher-end performance tended to be preferred by customers with larger budgets and higher requirements for camera, memory, and gaming capability. In contrast, budget-oriented brands were linked to customers prioritizing affordability over advanced specifications. The LDA canonical plot further supported these findings by showing visible separation between brand groups along LD1 and LD2, suggesting that the model successfully captured the major feature combinations that influence preference.

Variable-importance rankings also confirmed that some predictors have a much stronger role in distinguishing brands than others. Budget, camera quality, and RAM were the strongest contributors, illustrating that Lebanese customers tend to balance performance needs with economic constraints. Pairwise scatterplots reinforced this pattern by showing clusters of customers with similar characteristics gravitating toward the same device type.

Overall, the results demonstrate that discriminant analysis is an effective method for identifying which user characteristics and device features drive smartphone brand preference. These insights can support decision-making in retail environments by guiding stock selection, marketing strategies, and personalized recommendations. Future work could incorporate larger real datasets, new behavioral variables, or comparisons with other statistical learning models to further refine predictions and improve understanding of customer decision patterns.

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Appendix

Table 3. LDA Classification Accuracy

Dataset	Accuracy
Training	0.8933333
Testing	0.8000000

Table 4. Confusion Matrix for LDA Model

	iPhone	Other	Samsung
iPhone	7	0	1
Other	0	4	2
Samsung	2	0	9

Table 5. Coefficients of Linear Discriminants

	LD1	LD2
Age	0.0012553	0.0079698
Camera_Quality	-0.2852365	-0.4672362
Battery_Life	0.0340729	0.0305846
Screen_Size	-0.3874933	0.3790353
Memory_Storage	-0.0002732	-0.0005327
RAM	-0.0910563	0.1339830
Gaming_Usage	-0.1246136	0.0771008
Brand_Loyalty_Score	0.0835175	-0.0402823
Budget_USD	-0.0049558	0.0005576
SIM_TypeSIM	-0.4856793	0.3958034

Table 6. Group Means for Predictors by Brand

Preferred_Phone	Age	Camera_Quality	Battery_Life	Screen_Size	Memory_Storage	RAM	Gaming_Usage	Brand_Loyalty_Score	Budget_USD
iPhone	37.40000	8.600000	17.96000	6.220000	174.0800	6.960000	5.240000	5.400000	966.2800
Other	37.21053	8.473684	19.89474	6.084211	148.2105	5.789474	4.157895	5.578947	270.6316
Samsung	38.39286	6.517857	18.67857	6.232143	166.8571	7.107143	6.232143	4.696429	674.4821